AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 23: Minimal Residual and GMRES

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Outline

1. **Minimal Residual for Symmetric Systems (MC§11.4.1)**
2. Generalization to Nonsymmetric Systems
3. Convergence of GMRES

Review: CG Based on Lanczos Iteration

- In Lanczos iteration for $A \in \mathbb{R}^{n\times n}$, starting from $q_1 = b/\|b\|$

$$AQ_k = Q_{k+1}\tilde{T}_k, \tag{1}$$

  where $\tilde{T}_k$ is $(k+1)\times k$; Q_k is composed of orthonormal basis of $\mathcal{K}_k$
- Let $x_k = Q_k y_k$. Then,

$$r_k = b - Ax_k = b - AQ_k y_k = b - Q_{k+1}\tilde{T}_k y_k$$

- Let $Q_k^\top r_k = Q_k^\top\left(b - Q_{k+1}\tilde{T}_k y_k\right) = 0$ (i.e., $r_k \perp \mathcal{K}_k$), we obtain

$$Q_k^\top Q_{k+1}\tilde{T}_k y_k = Q_k^\top b$$

  where $Q_k^\top Q_{k+1}\tilde{T}_k = T_k$ and $Q_k^\top b = \beta e_1 \in \mathbb{R}^k$ with $\beta = \|b\|$
- Hence,

$$T_k y_k = \beta e_1 \tag{2}$$

  where $T_k = Q_k^\top A Q_k$ is tridiagonal, and is SPD if A is SPD
- It takes $\mathcal{O}(1)$ flops to update Cholesky factorization of T_k and then $\mathcal{O}(k)$ flops to solve (2)

How to Generalize CG to Symmetric Indefinite Systems?

- If A is not positive definite, Cholesky factorization of T breaks down
- What if we use LU or QR for T_k instead?
  - SYMMLQ uses QR factorization of T_k to solve tridiagonal system
  - SYMMLQ enforces $r_k \perp \mathcal{K}_k(A, b)$ like CG, but it does not minimize any specific norm in e_k, since $\|\cdot\|_A$ is no longer a proper norm
- Alternative approach is MINRES, which minimize 2-norm of residual at each step
- Both SYMMLQ and MINRES were developed by C. C. Paige and M. A. Saunders (1975)
- MINRES is equivalent to E. Stiefel's Conjugate Residual (CR, 1955) method for SPD systems, and D.G. Luenberger's extension of CR to indefinite systems (1969)

Minimal Residual Method (MINRES)

- Similar to CG, consider Lanczos iteration of A with $q_1 = b/\|b\|$

$$AQ_k = Q_{k+1}\tilde{T}_k$$

Let $x_k = Q_k y_k$. Then,

$$r_k = b - Ax_k = b - AQ_k y_k = b - Q_{k+1}\tilde{T}_k y_k$$

- Since $q_1 = b/\|b\|$, $b \in \text{range}(Q_{k+1})$. Therefore,

$$\|r_k\| = \|b - Q_{k+1}\tilde{T}_k y_k\| = \left\|Q_{k+1}^{\top}\left(b - Q_{k+1}\tilde{T}_k y_k\right)\right\| = \left\|\beta e_1 - \tilde{T}_k y_k\right\|,$$

where $Q_{k+1}^{\top} b = \beta e_1 \in \mathbb{R}^{k+1}$ with $\beta = \|b\|$

- At each step, we solve $(k+1) \times k$ least-squares problem

$$\tilde{T}_k y_k \approx \beta e_1 \tag{3}$$

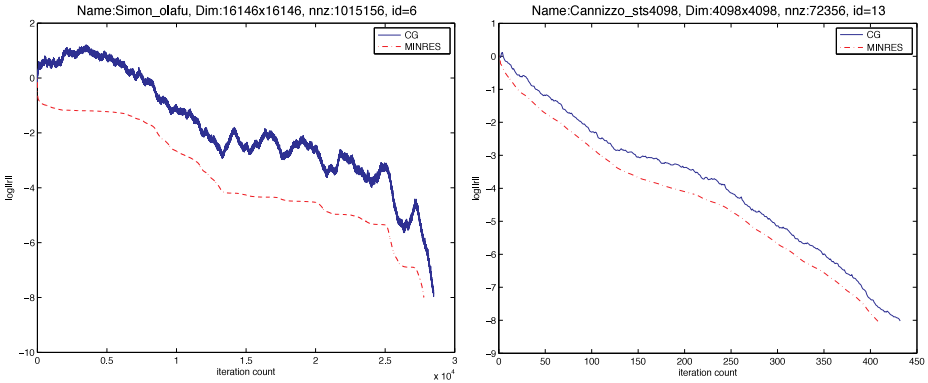
to minimizes $\|r_k\| = \left\|\tilde{T}_k y_k - \beta e_1\right\|$, where $\tilde{T}_k$ is tridiagonal.

- It takes $\mathcal{O}(1)$ flops to update QR of $\tilde{T}_k$ using Givens rotation and then $\mathcal{O}(k)$ flops to solve (3)

Comparison of CG, MINRES and SYMMLQ

	CG	SYMMLQ	MINRES
objective	$\min \|e_k\|_A$	$r_k \perp \mathcal{K}_k(A, b)$	$\min \|r_k\|_2$
internal solver	$T = LDL^\top$	$T = LQ$	$\tilde{T} = QR$
monotonicity ↘	$\|e_k\|_A$	-	$\|r_k\|_2$
flops per iteration	$8n$		$9n$

Comparison of CG, MINRES and SYMMLQ (Cont.)



Source: Fong, D. C. L., & Saunders, M. A. (2012). CG versus MINRES: An empirical comparison. *SQU Journal for Science*, 17(1), 44-62.

Convergence of MINRES

- If k_* is dimension of smallest invariant space that contains r_0, then MINRES terminates in k_* steps in exact arithmetic
- In particular, if A has s distinct eigenvalues, MINRES converges in $\leq s$ iterations.
- MINRES minimizes $\|p_k(A)r_0\|_2$ at kth step, with $r_0 = b$, where p_k is degree-k polynomial $p_k(x) = 1 + c_1 x + c_2 x^2 + \cdots + c_k x^k$
- Analogous to CG,

$$\frac{\|r_k\|_2}{\|r_0\|_2} \leq \inf_{p_k} \max_{\lambda} |p_k(\lambda)|$$

- If A is SPD, $\inf_{p_k} \max_\lambda |p_k(\lambda)| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \approx 2\left(1 - \frac{2}{\sqrt{\kappa}}\right)^k$ for large κ. Hence, MINRES also takes up to $\mathcal{O}\left(\sqrt{\kappa}\right)$ iterations
- If A is indefinite, MINRES may take up to $\mathcal{O}(\kappa)$ iterations in worst case (Benzi, Golub & Liesen (2005 *Acta Numerica*))

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Overview of Methods for Nonsymmetric Systems

- Various extensions of Krylov subspace methods have been proposed for nonsymmetric matrices (GMRES, BiCG, BiCGSTAB, QMR, etc.)
- Three different approaches
  - Generalization of minimal residual with Arnoldi iteration (GMRES, NLA§35, MC§11.4.3)
  - CG-style method with nonsymmetric Lanczos iteration (BiCG, NLA§39, MC§11.4.5)
  - Quasi minimal residual with nonsymmetric Lanczos iteration (QMR, NLA§39, MC§11.4.5)
- GMRES (Generalized Minimal RESidual) is one of most well known, developed by Saad and Schultz (1986)

Review: Arnoldi Iteration

- In Arnoldi iteration with $A \in \mathbb{R}^{n\times n}$ and $q_1 = b/\|b\|$,

$$AQ_k = Q_{k+1}\tilde{H}_k, \tag{4}$$

where $Q_k = [q_1 \mid q_2 \mid \cdots \mid q_k]$ is composed orthonormal basis of $\mathcal{K}_k(A, b)$, and $\tilde{H}_k$ is $(k+1)\times k$ upper Hessenberg

Algorithm: Arnoldi Iteration

given random nonzero b, let $q_1 = b/\|b\|$
for $k = 1, 2, 3, \ldots$
    $v = Aq_k$
    for $j = 1$ **to** k
        $h_{jk} = q_j^* v$
        $v = v - h_{jk} q_j$
    $h_{k+1,k} = \|v\|$
    $q_{k+1} = v/h_{k+1,k}$

This is essentially modified Gram-Schmidt orthogonalization of $\mathcal{K}_k(A, b)$

Derivation of GMRES

- Similar to MINRES, let $x_k = Q_k y_k$. Then,

$$r_k = b - Ax_k = b - AQ_k y_k = b - Q_{k+1}\tilde{H}_k y_k$$

- Since $q_1 = b/\|b\|$, $b \in \text{range}(Q_{k+1})$. Therefore,

$$\|r_k\| = \|b - Q_{k+1}\tilde{H}_k y_k\| = \left\|Q_{k+1}^{\top}\left(b - Q_{k+1}\tilde{H}_k y_k\right)\right\| = \left\|\beta e_1 - \tilde{H}_k y_k\right\|,$$

  where $Q_{k+1}^{\top} b = \beta e_1 \in \mathbb{R}^{k+1}$ with $\beta = \|b\|$

- At each step, we solve $(k+1) \times k$ least-squares problem

$$\tilde{H}_k y_k \approx \beta e_1 \tag{5}$$

  to minimizes $\|r_k\| = \left\|\tilde{H}_k y_k - \beta e_1\right\|$

High-Level GMRES Algorithm

Algorithm: GMRES

$q_1 = b/\|b\|$
for $k = 1, 2, 3, \ldots$
    Step k of Arnoldi iteration
    Update QR factorization of $\tilde{H}_k$
    Use QR to solve $\tilde{H}_k y_k \approx \beta e_1$
    $x_k = Q_k y_k$

- $\tilde{H}_k$ is upper Hessenberg. It takes $\mathcal{O}(k)$ flops to update QR factorization of $\tilde{H}_n$ using Givens rotation and $\mathcal{O}(k^2)$ flops to solve (5)
- $\|r_k\| = \left\|\tilde{H}_k y_k - \beta e_1\right\|$, so no need to compute explicitly from x_k
- However, Arnoldi iteration takes $\mathcal{O}(nk)$ flops, besides matrix-vector multiplication
- It is also possible to orthogonalize $\mathcal{K}_k(A, b)$ using Householder instead of Gram-Schmidt

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Convergence of GMRES for Diagonalizable Matrices

- Suppose A is diagonalizable. If k_* is dimension of smallest invariant space that contains r_0, then GMRES terminates in k_* steps in exact arithmetic. Here, $\mathcal{S}$ is *invariant* if $v \in \mathcal{S} \Rightarrow Av \in \mathcal{S}$
- $\|r_k\|_2$ decreases monotonically in GMRES
- GMRES minimizes $\|p_k(A)r_0\|_2$ at kth step, with $r_0 = b$, where p_k is degree-k polynomial $p_k(x) = 1 + c_1 x + c_2 x^2 + \cdots + c_k x^k$
- If $A = V\Lambda V^{-1}$, then $\|p_k(A)\| \leq \|V\|\|p_k(\Lambda)\|\|V^{-1}\| = \kappa(V)\|p_k(\Lambda)\|$
- Since $\|r_k\| = \|p_k(A)r_0\| \leq \|p_k(A)\|\|r_0\|$, we have

$$\frac{\|r_k\|_2}{\|r_0\|_2} \leq \inf_{p_k} \|p_k(A)\| \leq \kappa(V) \inf_{p_k} \|p_k(\Lambda)\| = \kappa(V) \inf_{p_k} \sup_{\lambda} |p_k(\lambda)|$$

- If A is not far from normal (i.e., $\kappa(V)$ is small), GMRES converges similarly to MINRES
  - If A is positive definite, then it may take $\mathcal{O}\left(\sqrt{\kappa}\right)$ steps
  - If A is indefinite, worst case is $\sim \mathcal{O}(\kappa)$ steps (this bound is not sharp)
- For nonnormal matrices, factor $\kappa(V)$ may be very pessimistic

Convergence of GMRES for General Matrices

- If A is not diagonalizable, if k_* is dimension of *smallest generalized invariant space* that contains r_0, then GMRES terminates in k_* steps in exact arithmetic. Here, $\mathcal{S}$ is *generalized invariant subspace* if $v \in \mathcal{S} \Rightarrow A^k v \in \mathcal{S}$ for $k \geq 0$, is related to generalized eigenvectors
- If $A = VJV^{-1}$, where J is the Jordan matrix, then

$$\|p_k(A)\| \leq \|V\|\|p_k(J)\|\|V^{-1}\| = \kappa(V)\|p_k(J)\|$$

and

$$\frac{\|r_k\|_2}{\|r_0\|_2} \leq \inf_{p_k} \|p_k(A)\| \leq \kappa(V) \inf_{p_k} \|p_k(J)\|$$

- Bound on $\|p_k(J)\|$ is more complicated to describe; see Tichý, Liesen, and Faber (2007)

Practical Considerations of GMRES

- In (full) GMRES, memory and computational cost both grow with k
- In practice, (full) GMRES is almost never used
- Instead, GMRES with restart, denoted by GMRES(k), is used, which runs GMRES for k steps, and then restart to solve residual equation
- Fundamental difficulties with restarted GMRES(k)
  - if k is too large, GMRES(k) is expensive in terms of cost per iteration
  - if k is too small, GMRES(k) may stagnate
- No practical guideline for optimal k for any input matrix
- Practical applications reply on preconditioners for faster convergence
- See Saad (2003), *Iterative Methods for Sparse Linear Systems*, 2nd Edition, SIAM, p. 179